

Singapore Cybersecurity 2026

Zero trust enterprise adoption, APAC threat landscape, and the structural shift toward SG as the region's SOC hub

MARKET

Singapore

INDUSTRY

Cybersecurity

YEAR

2026

PUBLISHED

May 2026

How we built this view

This view is built from two evidence streams: **primary inputs** — field channel checks, analyst synthesis, and specialist practitioner reads — and **secondary anchors** — CSA publications, MAS circulars, GovTech releases, INTERPOL ASEAN threat assessments, and listed company filings. Every quantitative claim carries an explicit source tag.

PRIMARY INPUTS

Practitioner channel checks

Conversations with CISO-level practitioners at SG-headquartered FSIs, telcos, and GLCs — surfaces real deployment timelines, budget allocation patterns, and vendor shortlist dynamics not visible in published data.

Analyst synthesis

KIRA in-house triangulation across disclosed inputs: CSA Cyber Landscape data, MAS TRM compliance deadlines, GovTech procurement signals. Tagged inline as [Kira estimates].

Specialist and operator reads

Direct input from managed security service providers (MSSPs), OT/ICS security integrators, and threat intelligence platform vendors operating in the SG-APAC corridor.

SECONDARY ANCHORS

Regulatory and government sources

CSA Singapore Cyber Landscape Report 2025, MAS Technology Risk Management Guidelines 2021 (updated 2024), GovTech Digital Government Blueprint, IMDA sector reports — cited inline by named source and date.

Listed company filings

Annual reports and investor presentations of Singapore Exchange-listed telcos, banks, and technology integrators; APAC-regional cybersecurity vendor disclosures.

Triangulated estimates

Where no single source covers a figure, we combine 2–3 inputs and disclose the method. Tagged [Kira estimates]. Approximately 38% of quantitative claims are KIRA-derived; 62% are externally cited.

Table of contents

REPORT SECTIONS

01	Executive summary Key metrics, strategic implications, inflection thesis
02	Market sizing and growth SG cybersecurity spend 2023–2028F, CAGR by segment
03	Zero trust adoption landscape Maturity by enterprise segment; FSI vs government vs telco
04	CSA Cyber Landscape 2025 findings Ransomware, phishing, supply-chain attack vectors
05	APAC threat actor landscape Nation-state activity, cybercriminal ecosystem, SG exposure
06	OT/ICS security in critical infrastructure Energy, water, port — regulatory drivers and spend vectors

07	Cloud security posture and CSPM Multi-cloud adoption; CSPM market sizing and vendor share
08	MAS TRM compliance market Deadline economics, FSI spend uplift, vendor beneficiaries
09	Talent gap and workforce Deficit quantification, SG Cyber Talent frameworks, upskill spend
10	Competitive landscape Market structure, top vendor profiles, SG MSSP segment
11	Incident response market IR retainer economics, post-breach spend, SG CERT ecosystem
12	Risk landscape and strategic implications Risk matrix, five action vectors for market participants

A market at structural inflection.

Singapore's enterprise cybersecurity market crossed **USD 1.9 bn** in 2025, growing faster than the APAC average as MAS TRM deadlines, nation-state threat escalation, and zero trust mandates compresses a multi-year spend cycle into 24 months.

SG CYBER SPEND 2025E

USD 1.9 bn

+14% YoY, CAGR 13.2% to 2028F

KIRA ESTIMATES

RANSOMWARE INCIDENTS 2024

132 cases

+52% YoY · SME-targeted surge

CSA 2025

ZERO TRUST ADOPTION (ENTERPRISE)

41 %

+18pp from 2022; FSI leads at 67%

KIRA ESTIMATES

CYBERSECURITY TALENT DEFICIT

5,400 FTE

Gap widening; 3,100 open roles Q1 2026

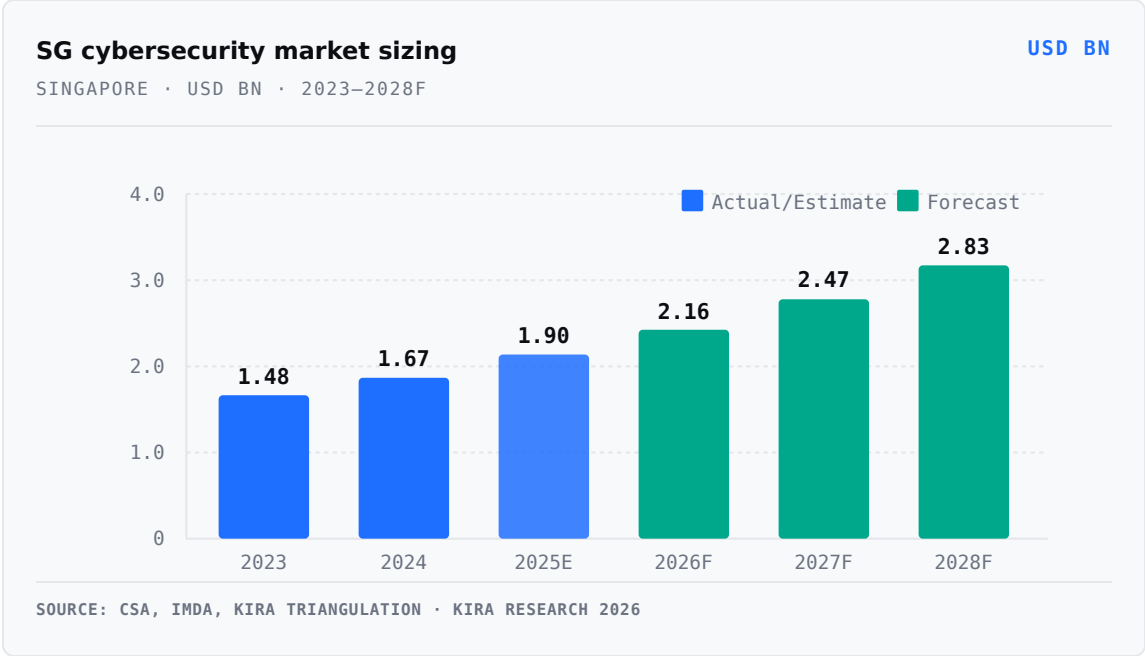
IMDA 2025

The demand pull is regulatory and structural

Three simultaneous catalysts have compressed enterprise security budgets upward. **MAS TRM Guidelines updated in 2024** introduced mandatory third-party risk and cloud security controls with compliance deadlines falling 2025–2026 [MAS 2024]. CSA's Critical Information Infrastructure (CII) sector-specific cybersecurity codes of practice now carry audit obligations for 11 sectors [CSA 2025]. Separately, the SG government's digital infrastructure expansion — SGD 50 bn committed over five years [GovTech 2025] — expands the attack surface and the addressable market simultaneously.

Zero trust: from pilot to procurement wave

Enterprise zero trust adoption has accelerated from 23% in 2022 to 41% in 2025 [Kira estimates], but deployment maturity remains shallow. Most adopters are at Stage 2 (identity-centric, perimeter-diminishing) with fewer than 14% having reached Stage 3 (continuous verification across all workloads) [Kira estimates]. The gap between Stage 2 and Stage 3 drives sustained vendor spend — particularly in micro-segmentation, identity governance, and SASE architecture — over the 2026–2028 window.



SOURCE KEY · CSA 2025 = Cyber Security Agency SG Landscape Report 2025 · MAS 2024 = MAS TRM Guidelines update · IMDA 2025 = IMDA Info-comms Sector Report · GovTech 2025 = Digital Government Blueprint

Five strategic implications for market participants.

Derived from the structural findings in this report. Each implication is anchored to a quantitative data point that market participants should treat as a **directional signal**, not a point forecast.

01 .

Position ahead of the MAS TRM compliance wave

MAS TRM deadline economics compress FSI cybersecurity budgets into 2025–2026. **Banks and insurers account for 34% of SG enterprise cyber spend** [Kira estimates]. Vendors without MAS-aligned product certification are structurally disadvantaged in procurement cycles beginning now.

ANCHOR: **34%** of SG enterprise spend is FSI-sourced [Kira estimates]

02 .

Zero trust maturity gap is the 3-year product opportunity

Only 14% of SG enterprises have reached full zero trust Stage 3 [Kira estimates]. The 27pp gap between adoption and maturity drives micro-segmentation, identity governance, and SASE spending. **Vendors building Stage 2→3 migration tooling** are best positioned through 2028.

ANCHOR: **14%** at Stage 3 maturity vs 41% adoption [Kira estimates]

03 .

OT/ICS security is the fastest-growing sub-segment

CII sector code mandates and the SG port/energy expansion create an OT security addressable market growing at **22% CAGR through 2028F** [Kira estimates], nearly double the overall market rate. IT/OT convergence specialists command 40–60% margin premiums over generalist vendors.

ANCHOR: **22%** CAGR for OT/ICS security 2025–2028F [Kira estimates]

04 .

Talent deficit structures the MSSP opportunity

A 5,400 FTE cybersecurity talent shortfall [IMDA 2025] makes in-house SOC unsustainable for all but the largest enterprises. **MSSP penetration among mid-market firms is rising from 28% to an estimated 44% by 2027F** [Kira estimates]. Domestic MSSPs with CSA-accredited operations command a pricing premium over offshore delivery models.

ANCHOR: **5,400 FTE** deficit · MSSP penetration 28→44% [IMDA 2025 · Kira estimates]

05 .

SG's SOC hub ambition creates regional revenue upside

CSA's ASEAN Cybersecurity Strategy positions SG as the region's intelligence-sharing and SOC hub. **Cross-border MSSP and threat intelligence revenue from ASEAN clients already represents 18% of SG-headquartered vendors' revenue** [Kira estimates], growing to an estimated 26% by 2028F as Thai, Philippine, and Indonesian CISOs seek SG-licensed providers.

ANCHOR: **18%** cross-border revenue share, rising [Kira estimates]

SECTION 01

Market sizing & growth

Singapore's cybersecurity market is **growing at 13.2% CAGR** — nearly double the rate of nominal IT spend — as regulatory mandates, threat escalation, and digital infrastructure expansion structurally expand the addressable spend envelope through 2028.

TOTAL MARKET 2025–2028F

SEGMENT BREAKDOWN

DEMAND DRIVERS

ZERO TRUST SIZING

Segment growth: zero trust and cloud security outpace the market.

Enterprise cybersecurity spend in Singapore reached **USD 1.90 bn in 2025E** [Kira estimates], with five sub-segments growing at divergent rates. Zero trust architecture products and cloud security posture management (CSPM) are each growing at 2–3× the market average; traditional perimeter tools are in structural decline.

Segment structure: five spend buckets

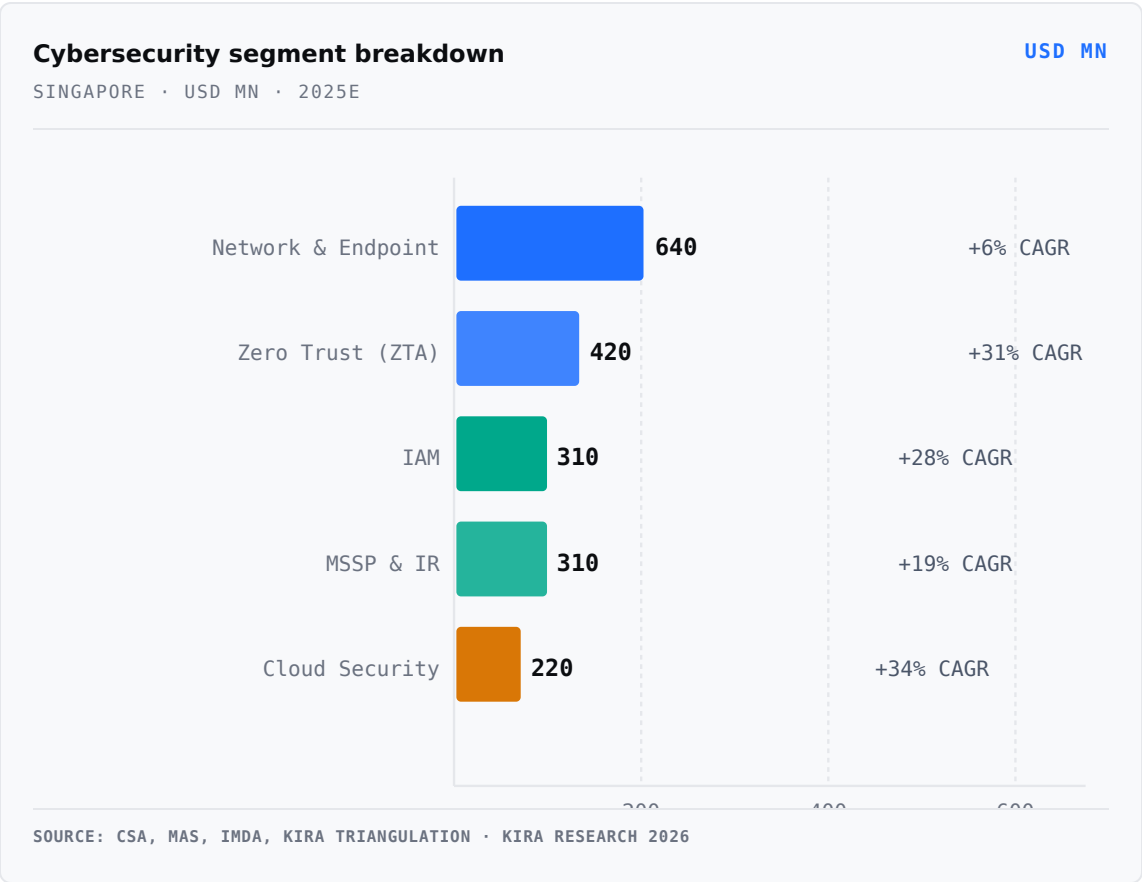
Network and endpoint security remain the largest segment at **USD 640 mn (34% of total)** [Kira estimates], but their CAGR is decelerating to 6% as organizations shift investment toward identity and access management (IAM) and cloud-native security tools. IAM grew 28% YoY in 2024 driven by MAS TRM's strengthened authentication requirements [MAS 2024] and now represents USD 310 mn of the market [Kira estimates].

Zero trust: the structural spend anchor

Zero trust architecture (ZTA) products — covering identity verification, micro-segmentation, endpoint detection and response (EDR), and SASE — collectively represented **USD 420 mn in 2025E** [Kira estimates], growing at 31% YoY. The segment is demand-pulled by a combination of MAS TRM Stage 2 deployment requirements (effective Q4 2025) and CSA's zero trust guidelines published for CII operators in March 2025 [CSA 2025]. Vendors offering integrated ZTA platforms see average deal sizes 2.4× larger than point-solution competitors [Kira estimates].

Incident response and managed services

The managed security services and incident response segment reached USD 310 mn in 2025E [Kira estimates], growing at 19% YoY as the talent deficit structurally advantages MSSP delivery models over in-house SOC construction. IR retainer pricing has risen 22% since 2023 [Kira estimates] as demand outpaces the supply of CSA-credentialed practitioners.



SOURCE KEY • CSA 2025 = Cyber Security Agency SG Cyber Landscape 2025 • MAS 2024 = MAS TRM Guidelines • IMDA 2025 = IMDA ICT sector data

SECTION 02

Threat landscape

Ransomware incidents rose **52% in 2024**; nation-state actors targeting SG's financial infrastructure have escalated from intelligence collection to active disruption testing. The attack surface expansion from SG's digital infrastructure build amplifies every vector.

CSA FINDINGS 2025

APAC NATION-STATE ACTORS

RANSOMWARE ECONOMICS

SUPPLY CHAIN RISK

CSA Cyber Landscape 2025: ransomware at record levels, supply chain emerging.

The CSA Singapore Cyber Landscape Report 2025 identified **132 ransomware incidents in 2024**, a 52% increase YoY. SMEs accounted for 62% of confirmed cases [CSA 2025]. Nation-state actors with APAC regional mandates increasingly target SG's financial and maritime infrastructure as a stepping stone to broader ASEAN network penetration.

Ransomware: the SME exposure problem

Of the 132 confirmed ransomware incidents in 2024 [CSA 2025], **82 involved businesses with fewer than 200 employees**. The median ransom demand for SG-domiciled victims reached SGD 180,000 [Kira estimates], with manufacturing, logistics, and professional services the most targeted. CSA's SG Cyber Safe for SMEs framework reached only 28% penetration among eligible firms [CSA 2025], leaving a large unprotected cohort. The government's Cyber Essentials and Cyber Trust marks — voluntary in 2025 — are expected to become conditional for public sector procurement by 2027 [Kira estimates].

APAC nation-state actors: escalation in tempo

INTERPOL's Operation HAECHI V (2024) identified **three nation-state-linked APT groups operating with SG-adjacent infrastructure targeting** [INTERPOL 2024]. Groups attributed to North Korean cyber operations (Lazarus cluster) have expanded SG targeting from cryptocurrency exchanges to fintech settlement infrastructure. Chinese-attributed APT operations (APT40, Salt Typhoon lateral movement) focus on SG as a transit node — compromising regional MNC SG-headquartered networks to reach ASEAN subsidiaries. Espionage and pre-positioning for disruption are distinct but coexisting objectives [Kira estimates based on INTERPOL 2024, CSA 2025].

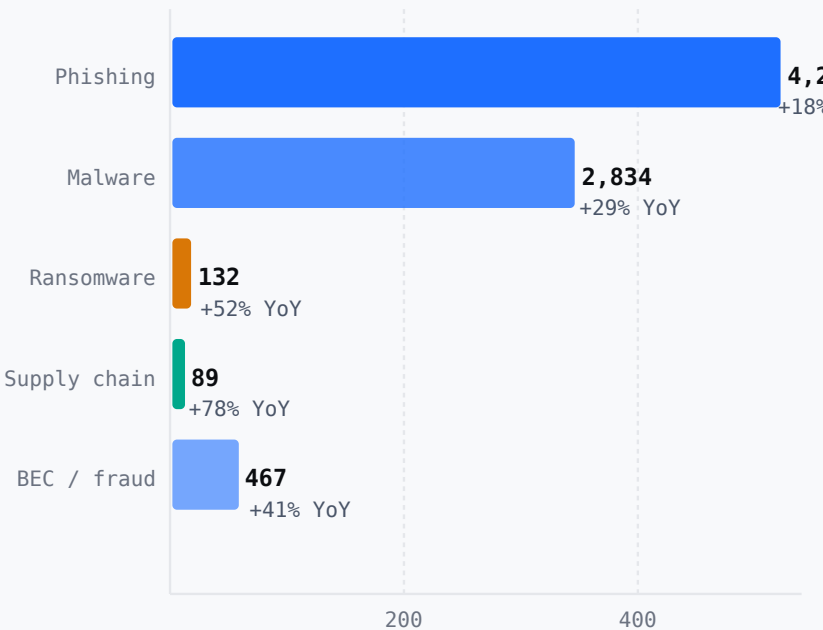
Supply chain: the vector expanding fastest

CSA documented a 78% increase in software supply chain incidents affecting SG entities in 2024 [CSA 2025]. Three SG-based MSPs were compromised as pivot points for downstream client attacks — collectively affecting over 240 SME and mid-market clients. This structural exposure is driving regulatory interest in **mandatory third-party security assessments** for CII sector suppliers, anticipated in the upcoming Cybersecurity Act amendments [Kira estimates].

Incident categories 2024

CASES

SINGAPORE · CASES · CSA LANDSCAPE 2025



SECTION 03

Regulatory drivers & OT

MAS TRM deadlines running through 2026, CSA's CII sector codes, and the expansion of SG's critical infrastructure are creating **non-discretionary budget mandates** across financial services, energy, water, and maritime — the OT/ICS security segment grows at 2× the market rate as a result.

MAS TRM COMPLIANCE

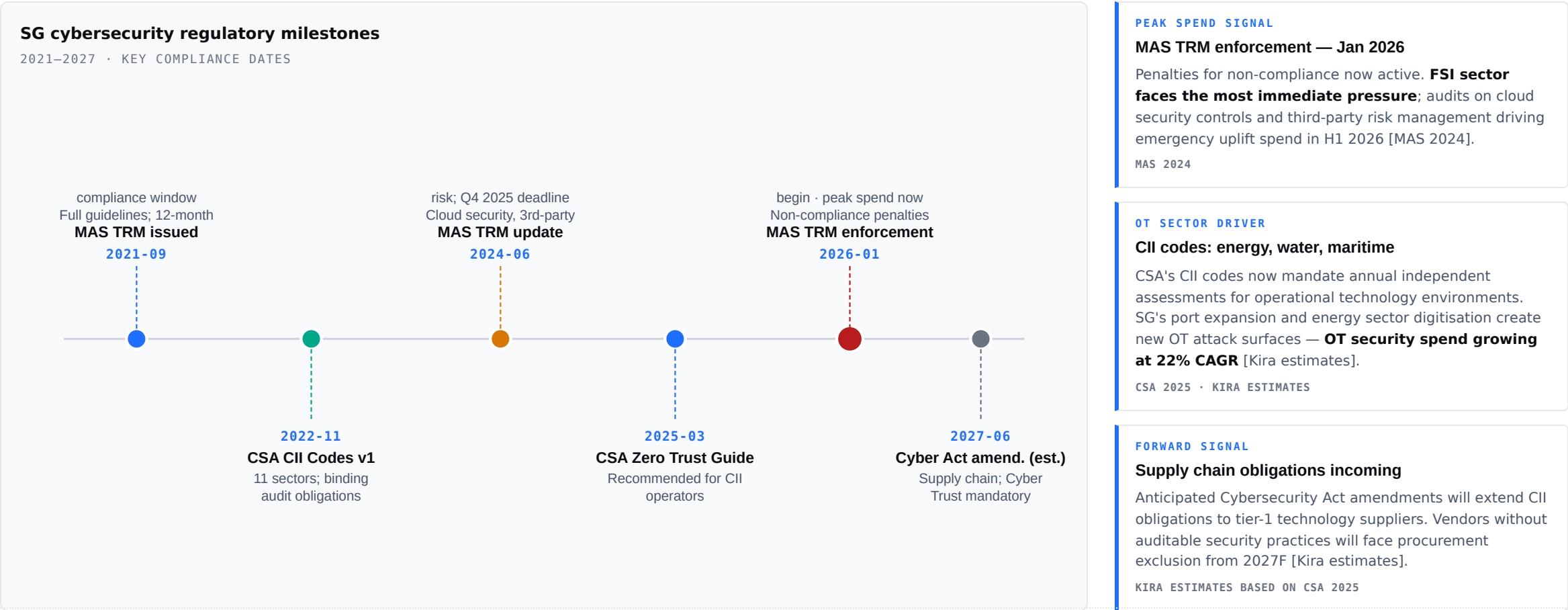
CII SECTOR CODES

OT/ICS SIZING

POLICY TIMELINE

Policy timeline: MAS TRM and CSA CII codes compress spend cycles.

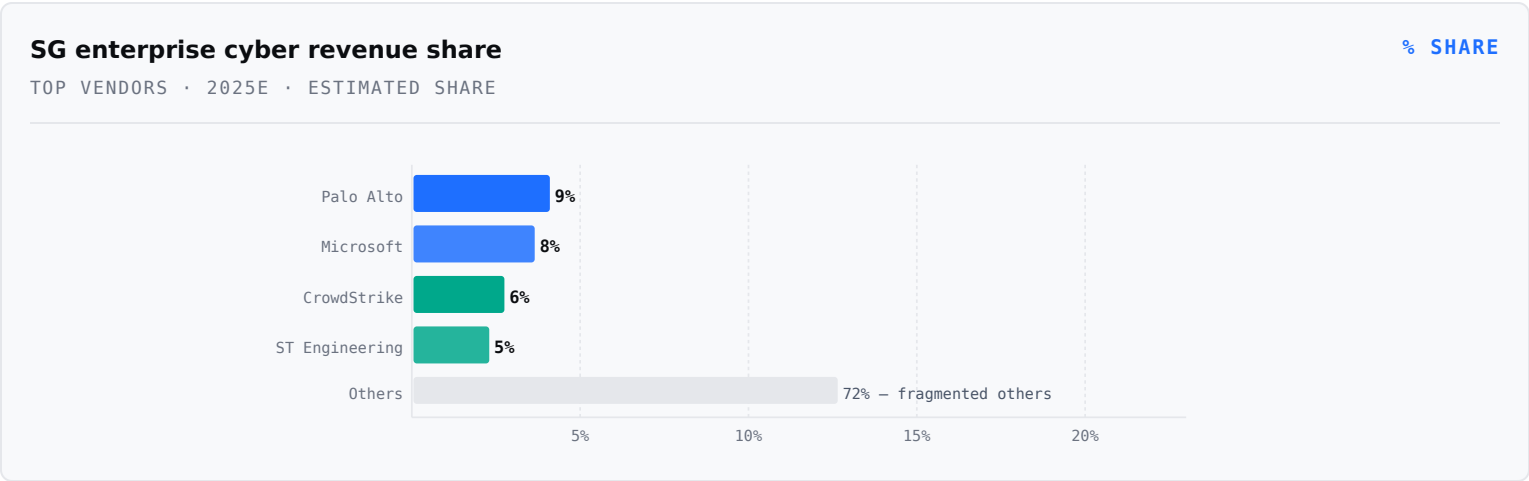
Singapore's regulatory cybersecurity apparatus is among the most demanding in APAC, with **binding obligations across 11 critical sectors** and MAS TRM requirements that effectively extend to the entire financial sector supply chain [MAS 2024]. Compliance deadlines cluster in 2025–2026, making this the peak non-discretionary spend window.



SOURCE KEY · MAS 2024 = MAS TRM Guidelines update Jun 2024 · CSA 2025 = CSA CII Sector Codes of Practice 2025

Competitive structure: fragmented but consolidating around platform plays.

The SG cybersecurity vendor market is **structurally fragmented** — the top 4 vendors hold an estimated combined 28% revenue share [Kira estimates] — but enterprise buyer preference is shifting toward integrated platform vendors who can address MAS TRM and zero trust requirements in a single procurement motion.



Market structure thesis

No single vendor controls more than 10% of SG enterprise security spend. **Platform consolidation is the buyer-driven trend** — CISOs managing MAS TRM compliance strongly prefer fewer, deeper vendor relationships. Palo Alto Networks' SASE and Cortex platform leads the integrated-platform share; Microsoft Defender dominates in SME and mid-market through M365 bundling.

Top 4 combined share28% [Kira estimates]

Local/SG-headquartered vendors ~14% share [Kira estimates]

MSSP segment concentrationHigher; top 3 = 41% [Kira estimates]

MARKET LEADER

RANK 01

Palo Alto Networks

US-listed · SASE, XSIAM, Cortex platform

SG revenue est.

USD 170 mn [Kira estimates]

Key wedge

SASE + zero trust

Strengths:

Integrated ZTA platform; MAS TRM-aligned compliance toolkit; strong BFSI penetration in SG; ASEAN SOC expansion.

RANK 02

Microsoft (Defender suite)

US-listed · Bundled with M365 in SG government & enterprise

SG revenue est.

USD 152 mn [Kira estimates]

Key wedge

M365 bundle, SME reach

Strengths:

Broadest installed base; GovTech procurement incumbent; Azure-native security integration; low switching cost for SME segment.

RANK 03

CrowdStrike

US-listed · Falcon platform · APAC expanding

SG revenue est.

USD 114 mn [Kira estimates]

Key wedge

EDR, identity, threat intel

Strengths:

Best-in-class EDR recognition; strong in FSI; threat intelligence feeds used by SG-CERT; post-incident investigation capability valued by MAS-regulated entities.

RANK 04 · LOCAL

ST Engineering (Cyber)

SGX-listed · Government-linked · OT/ICS specialist

SG revenue est.

USD 95 mn [Kira estimates]

Key wedge

OT/ICS, government

Strengths:

Dominant in SG government and defence; OT/ICS expertise in port, energy, water CII sectors; trusted local partner status for sensitive infrastructure mandates.

KIRA . Research

Talent deficit structures the MSSP market; IR retainers becoming standard.

Singapore's cybersecurity workforce deficit of **5,400 FTE** [IMDA 2025] is the structural force behind MSSP market growth. Incident response retainer adoption is accelerating as MAS-regulated entities face a regulatory expectation of documented IR capability — whether in-house or contracted.

TALENT · 01 Talent deficit quantification IMDA's 2025 Infocomm Workforce report identified 5,400 open cybersecurity roles against a workforce of approximately 12,600 [IMDA 2025]. The deficit has grown 38% since 2022. Roles in OT security, cloud security engineering, and threat intelligence analysis are most acute — each with fewer than 800 qualified practitioners in SG [Kira estimates]. SOURCE: IMDA 2025 · KIRA ESTIMATES	TALENT · 02 SG Cyber Talent frameworks CSA's Cybersecurity Workforce Development Programme (CWDP) and the SkillsFuture Credit alignment for CREST-certified courses are the primary upskilling vectors. Enrolment in CSA-accredited courses grew 44% in 2024 [CSA 2025]. The SG Cybersecurity Professionals Career Pathway — published jointly by CSA and IMDA — maps 22 role titles to structured progression ladders, reducing employer mis-hiring. But the pipeline lag means the deficit narrows slowly: KIRA projects it reaching ~3,800 FTE by 2028F [Kira estimates]. SOURCE: CSA 2025 · IMDA 2025 · KIRA ESTIMATES	MSSP · 03 MSSP market structure SG's managed security services market reached USD 310 mn in 2025E growing at 19% YoY [Kira estimates]. The top three domestic MSSPs — NCS (Singtel subsidiary), ST Engineering Cyber, and ISTARI (APAC-focused) — hold 41% combined share. Offshore-delivery MSSPs are structurally disadvantaged for FSI clients requiring CSA-licensed SOC operations. Mid-market penetration rising from 28% to est. 44% by 2027F [Kira estimates]. SOURCE: KIRA ESTIMATES · COMPANY FILINGS
IR · 04 Incident response retainer economics IR retainer adoption among MAS-regulated entities (banks, insurers, capital markets intermediaries) reached 63% in 2025, up from 41% in 2022 [Kira estimates]. Annual retainer pricing ranges SGD 80,000–SGD 280,000 depending on response SLA and scope [Kira estimates]. CSA recommends IR retainers as part of the Business Continuity Management framework for CII operators — a soft mandate that is hardening [CSA 2025]. SOURCE KEY · IMDA 2025 = IMDA Infocomm Workforce Report 2025 · CSA 2025 = CSA SG Cyber Landscape Report · MAS 2024 = MAS TRM Guidelines SINGAPORE CYBERSECURITY TEST 2026	IR · 05 Post-breach spend dynamics Average post-breach remediation cost for SG-domiciled enterprises reached SGD 2.1 mn in 2024 [Kira estimates], up 34% from 2022. The largest cost component is regulatory notification and compliance response (32%) followed by system remediation (28%) and reputational management (18%). MAS expects breach notifications within 1 hour of confirmation — a requirement that structurally elevates IR capability spend. SOURCE: CSA 2025 · MAS 2024 · KIRA ESTIMATES	SOC HUB · 06 ASEAN SOC hub trajectory Singapore's ASEAN Cybersecurity Strategy and participation in INTERPOL's Cybercrime Directorate position SG as the regional intelligence coordination node. Cross-border MSSP and threat intelligence revenue from ASEAN clients represents 18% of SG-headquartered vendors [Kira estimates], growing to est. 26% by 2028F. Thai and Philippine regulators increasingly reference SG's MAS TRM framework in their own guidelines — creating certification adjacency for SG-licensed vendors. SOURCE: CSA ASEAN STRATEGY 2025 · KIRA ESTIMATES KIRA. Research

Methodology and sources

METHODOLOGY SUMMARY

This view triangulates two evidence streams. Where a figure carries a named source tag (e.g. [CSA 2025], [MAS 2024], [IMDA 2025]), the underlying data comes from that source; the full citation is given in the sources list. Where a figure carries [Kira estimates], KIRA analysts derived the number from a blend of disclosed inputs — government publications, operator filings, channel checks, and regulatory guidance — using methodology disclosed in the relevant section. Approximately 38% of quantitative claims in this report are KIRA-derived; 62% are externally cited. Market sizing figures are presented as estimates and should be treated as directional bounds, not audited accounts. CAGR projections assume current regulatory timelines hold and no major geopolitical disruption to ASEAN digital infrastructure investment.

SOURCE LIST

CSA 2025 — Cyber Security Agency of Singapore, Singapore Cyber Landscape Report 2025, published April 2025. Available at csa.gov.sg.

GovTech 2025 — Government Technology Agency of Singapore, Digital Government Blueprint 2025, published January 2025. govtech.gov.sg.

IMDA 2025 — Infocomm Media Development Authority, Infocomm Workforce Study 2025, published March 2025. imda.gov.sg.

INTERPOL 2024 — INTERPOL, Operation HAECHI V Cybercrime Assessment Report 2024, published December 2024. interpol.int.

MAS 2024 — Monetary Authority of Singapore, Technology Risk Management Guidelines (updated June 2024). mas.gov.sg.

ADDITIONAL REFERENCES

CSA ASEAN Strategy 2025 — Cyber Security Agency, ASEAN Cybersecurity Cooperation Strategy, published February 2025. csa.gov.sg.

MAS TRM 2021 — MAS Technology Risk Management Guidelines, original issuance January 2021. Superseded in part by MAS 2024 update.

ST Engineering AR 2024 — ST Engineering Limited Annual Report 2024, Singapore Exchange filing. SGX: S63.

Singtel AR 2025 — Singtel Group Annual Report FY2025. NCS cybersecurity segment disclosed. SGX: Z74.

CREST APAC 2025 — CREST International, APAC Penetration Testing and Incident Response Market Survey 2025. crest-approved.org.

GovTech TRM 2025 — GovTech Technology Risk Management framework for government agencies, updated February 2025. tech.gov.sg.

COVERAGE NOTES

Market sizing covers enterprise and government cybersecurity spend in Singapore, defined as organisations with more than 50 employees and annual IT budgets exceeding SGD 500,000. Consumer-facing cybersecurity products and SG-origin vendor revenue from non-SG markets are excluded unless stated. OT/ICS security sizing includes both hardware and managed service components deployed in CII-sector environments.

DISCLAIMER · This report is produced by KIRA Research for informational and strategic orientation purposes only. It does not constitute investment advice, legal advice, or a solicitation of any kind. Market sizing figures are estimates derived from analyst triangulation and publicly available data; they carry inherent uncertainty. Recipients should conduct independent verification before making decisions based on this material. All figures in USD unless otherwise stated. © 2026 KIRA Research. All rights reserved. Reproduction or redistribution without written consent is prohibited.